Event-Triggered Link-Quality and Topology Prediction for Mobile Underwater Wireless Sensor Networks: A Kalman Filter and Reinforcement Learning Approach

Xuefei Ma, Ghulam Ali, Zhongwei Shen, Insha Hassan, Muhammad Aon, Qingling Liu, Haifeng Zhu, and Ying Cui

***Abstract*—Mobile underwater wireless sensor networks must keep link quality current as vehicles move, yet a fixed neighbor-broadcast timer spends scarce acoustic airtime while the channel is temporarily stable and can still miss the instants at which a link-quality forecast has diverged from the channel. This article presents Event-Triggered Link-Quality and Topology Prediction (E-LQTP), an extension of the link-quality-aware routing protocol LQARP. A dual-track extended Kalman filter maintains a measurement-corrected relative-motion estimate and a forecast of the same state that receives no measurement update. Their innovation, gated by a persistence test and a variance cap, emits a HELLO only when that forecast diverges from a per-tick channel query. E-LQTP inherits LQARP's multipath channel model, depth-progressive candidate set, and Bellhop integration; the Q-learning reward is left unchanged at the validated operating point. On the same Western Pacific site, with forty relays over six hundred seconds and five seeds, every protocol reads the same live pose. E-LQTP then matches LQARP's packet delivery, hops, delay, and energy per data bit, while cutting control traffic to one-third of a five-second HELLO grid under two-metre-per-second current drift and to about one-half under a Rankine eddy. Additive predictive reward terms do not improve the data plane. The control-plane gain is the event-triggered HELLO under that shared pose, with the reward left unchanged.**

***Index Terms*—Event-triggered communication, extended Kalman filter (EKF), link quality-aware routing, mobile underwater wireless sensor network (UWSN), Q-learning, underwater acoustic channel.**

Note to Practitioners: A mobile AUV mesh that copies a fixed HELLO timer spends acoustic airtime on a schedule that ignores whether a link-quality forecast has diverged from the channel. E-LQTP keeps LQARP's Bellhop lookup, depth-progressive candidate set, and Q-learning reward, and fires a 400-bit HELLO only on a rising edge of a persistence-gated dual-track innovation. On the Western Pacific site ($N = 40$), with every protocol reading the same live pose, the validated point (trigger on, $\beta_1 = \beta_2 = 0$) matches LQARP's data plane while cutting control traffic to $0.33\times$ a $5\,\mathrm{s}$ grid under $2\,\mathrm{m/s}$ drift and to $0.51\times$ under a Rankine eddy. Leave the reward weights at zero. Energy per bit follows LQARP and excludes HELLO joules; those joules are reported as control overhead.

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I. Introduction

THE Internet of Underwater Things (IoUT) deploys acoustic multi-hop meshes for environmental observation, resource exploration, and maritime awareness [1]–[4]. Underwater wireless sensor networks (UWSNs) must route under scarce bandwidth, long delay, and a time-varying multipath channel [1], [3], [5]. An increasingly important case is the *mobile* UWSN, in which autonomous underwater vehicles (AUVs) and unmanned surface vehicles (USVs) assist monitoring, adaptive sampling, and mobile relay meshes [2], [6]–[8]. There the static or moored-relay assumption common in prior routing work—including LQARP—breaks down, because relative motion continuously reshapes geometry and instantaneous link quality [9], [10].

Routing under mobility is not new, yet two threads remain poorly aligned with the control-plane cost of keeping link-quality state accurate while vehicles move. First, Q-learning and multi-agent reinforcement-learning UWSN protocols typically assume that neighbor and link state can be refreshed by periodic beaconing, or they omit an explicit relative-motion model that would tell a node *when* advertised link state has become stale [10]–[14]. Periodic HELLO traffic then consumes scarce acoustic bandwidth even while the channel is temporarily stable. Second, event-triggered Kalman filtering now schedules filter packets for IoUT target tracking and remote estimation [15]–[18], Kalman SNR prediction is used to adapt a modem [19], and event beacons appear in underwater named data networking [20], [21]. Those designs schedule tracking packets, modem updates, or named-data events. The subsections below identify which of them, if any, already fire a routing HELLO from an SNR forecast that is withheld from the per-tick channel query. Motivated by these challenges, we review existing reinforcement-learning underwater routing and event-triggered estimation, and identify why neither, on its own, resolves that control-overhead cost.

A. Related Work

E-LQTP inherits LQARP's multipath channel model, candidate-set construction, and Bellhop/Aqua-Sim integration unmodified, so this section does not reopen the simulation-tool argument treated in that work [9]. The subsections that

follow isolate the gaps that mobility introduces on top of that foundation.

1) Reinforcement-Learning UWSN Routing: Nandyala, Kim, and Cho propose QTAR, a Q-learning topology-aware protocol that selects a next-forwarder from a topology-constrained candidate set and uses implicit cut-vertex recognition to avoid energy-wasting detours [11]. Zhang, Chen, and Xie later add service-aware rewards and a holding time for opportunistic forwarding (SAQR) [12]. Yuan *et al.* nest Q-learning inside unequal clusters [22], and Wang *et al.* accelerate Q-table convergence with a depth-aware opportunistic reward [23], [24]. Mohanraju and Lokam replace the tabular update with a deep Q-network that scores residual energy, depth, and node degree (ADRP-DQL) [14]. Wang *et al.* further split the reward by value of information so that urgent and delay-tolerant packets share the same MARL forwarder (MARV) [25]. Liu *et al.* design CLORP, coupling multi-agent reinforcement learning with opportunistic forwarding and an adaptive learning rate [13]. Xu *et al.* introduce STGR for multi-UUV meshes: a distributed spatial-temporal graph records neighbor relations, link quality, and connectivity duration over periodic intervals, and Q-learning then selects a forwarder on that graph [10]. He *et al.* fold a trust score into the same opportunistic Q-learning loop so that malicious or unstable relays are down-ranked [26]. Li *et al.* assist the Q-update from an SDN controller (SDN-QLTR) [27]. LQARP itself folds Bellhop SNR, residual energy, delay, and neighbor count into a Q-learning reward on a stationary Western Pacific mesh [9]. Its data plane is the parent of E-LQTP; it does not address when a mobile node must re-advertise that reward state. Li, Wu, Zhu, and Ren come closest to the control rule studied here. Their Q-learning link-aware protocol predicts neighbor motion with an extended Kalman filter and sets the HELLO period to a fraction of the shortest predicted link expiration time, $T_H(N_i) = \tau \min_j T_{\mathrm{LET}}(N_i, N_j)$ with $\tau \in (0, 1)$ [28]. High relative motion shortens the period and low relative motion lengthens it, and the same expiration time enters the reward through an entropy-weight combination. That rule already withholds some beacon traffic while a geometrically predicted link is expected to survive. It remains a periodic timer scaled by link lifetime, and it spends the predicted lifetime inside the reward. It does not compare an SNR forecast, withheld from measurement updates since the last velocity write, with a channel query taken every tick. In the other protocols above, neighbor perception remains a periodic exchange or a timer-rebuilt graph.

2) Mobility-Aware and AUV-Assisted Collection: Wang *et al.* plan cooperative multi-AUV collection with local–global deep Q-learning and a separate emergency multi-hop route [6]. Liu *et al.* and Lu and Zuo treat the AUV as a path-planning agent whose reward includes ocean current and communication capacity [7], [8]. Nandyala and Cho dispatch an AUV to prevent isolated sub-networks once isolation is declared [29]. Jiang *et al.* opportunistically switch between acoustic and radio hops in a cooperative surface–subsurface mesh [30]. Yuan *et al.* jointly optimize multi-objective routes and a centralized cross-layer schedule of power, links, and next hops [31], [32]. In all of these designs the vehicle is a collector, a repair asset, or a radio relay—not a source of estimated relative motion that could gate neighbor broadcasts before advertised SNR goes stale.

3) Opportunistic, Clustered, and Information-Centric Forwarding: Jiang, Han, and Lin survey opportunistic IoUT routing and note that candidate-set maintenance is still a periodic control cost [4]. Zhu *et al.* cluster relays and predict voids with a Markov model (RCRP) [33], later adding a GMM–HMM–LSTM adaptive route [34]; a companion COMST article gives trust-management guidelines for the same setting [5]. Lee *et al.* form energy-efficient clusters with a fire-hawk optimizer and still emit periodic HELLO packets during neighbor discovery [35]. Wang *et al.* replace a passive interest with an actively forwarded beacon when a critical event occurs in underwater named data networking [20]. Li *et al.* avoid collisions among those NDN paths (RECAR) [21], and Wang, Lu, and Gao cache push and pull modes for information-centric underwater IoT [36]. Zhao *et al.* add time-reversal opportunistic forwarding [37]. Tian *et al.* hide source location behind layered candidates [38], and Yuan *et al.* validate a source-based disjoint multipath protocol in Zhoushan sea trials [39]. Those beacons and candidate sets are application-level or path-diversity events, not a routing HELLO tied to a relative-motion residual.

4) Event-Triggered Estimation and Channel Prediction: Priel and Zelazo develop event-triggered consensus Kalman filtering for time-varying networks with intermittent observations [16]. Sebastián *et al.* certify an event-triggered distributed Kalman filter under unknown correlations [17]. Deng and Xiong infer the remote state when the trigger decision itself is hidden [18], [40], and Zhong *et al.* schedule multiple sensors over packet-dropping links [41]. Closest in venue and in the use of a predicted state as the trigger reference, Li *et al.* design two media-specific event triggers for consensus Kalman filtering over a hybrid IoUT [15]. When a surface node does not transmit, neighbors propagate the last broadcast through the Kalman predict step, yielding a predicted information pair against which the next innovation is tested. That construction is the nearest cousin of the forecast track in Section III: both ask what a neighbor still believes after a missed transmission. Li *et al.* reset the predicted information pair when a packet arrives. The forecast in Section III is reset when the local broadcast flag is high, by copying the neighbor's current odometry, and the copy is not a demodulated packet. The scheduled object in [15] remains a filter packet for AUV *tracking*, rather than a HELLO that refreshes advertised link quality for Q-learning forwarding.

A related physical-layer thread predicts SNR itself rather than scheduling a packet. Busacca *et al.* compare N-state Markov and hidden-Markov models with a scalar Kalman filter on NATO LOON21 traces [19]. On those fixed, shallow links the Kalman mean absolute SNR error is $4.81\,\mathrm{dB}$ on the 11 November 2021 traces, $3.47\,\mathrm{dB}$ when that model is tested on 14 November, and $3.24\,\mathrm{dB}$ when it is tested on 18 November. The $2\,\mathrm{dB}$ cap in Section III is a design choice of this paper, placed below those three fixed-link errors. Busacca *et al.* use the filter to adapt a constellation; they do not prescribe a HELLO threshold. The dual-track EKF below is a

staleness detector for HELLO, not an SNR-RMSE competitor.

Overall, two issues remain open. First, the reinforcement-learning and opportunistic protocols reviewed above, other than QLAR, keep link state current by periodic beaconing or by a timer-rebuilt graph [9]–[11], [33]. QLAR does test predicted link lifetime, and it uses that lifetime both to scale a periodic HELLO and to shape the reward [28]. Second, event-triggered Kalman filtering schedules filter packets for IoUT tracking and remote estimation [15], [17], [41], Kalman SNR prediction adapts a modem [19], and event beacons appear in UNDN [20]. None of those three threads schedules a routing HELLO from the divergence between an SNR forecast that receives no measurement update and a channel measurement taken at every tick.

B. Contributions

The contributions of this article are summarized as follows.

1) To answer the two gaps jointly, we design a dual-track relative-motion estimator—a measurement-corrected track and a forecast track that receives no measurement update—so that an event trigger can be computed even though the channel-quality measurement is available every tick. Its innovation, gated by a persistence condition and a variance-capped threshold, decides when a node broadcasts the velocity copy that resets the forecast. The resulting Event-Triggered Link-Quality and Topology Prediction protocol (E-LQTP) replaces periodic HELLO traffic with that motion-aware copy.
2) We investigate whether the same relative-motion estimate can directly inform the Q-learning reward through a predictive link-quality term and a drift-aware forwarding term, and we characterize—rather than tune away—the conditions under which this does *not* improve routing: a bit-error-rate floor that creates a scale mismatch with an additive prediction penalty, and a radial-velocity sign persistence shorter than one hop's transit time. This is reported as a scoped, evidence-based negative result.

C. Paper Organization

The rest of this article is organized as follows. Section II introduces the extended system model of a mobile UWSN under uniform current drift and Rankine-vortex eddy motion. Section III details the design of E-LQTP, including the dual-track extended Kalman filter, the persistence-gated event trigger, and the scoped Q-learning reward terms. Section IV presents simulation results on the Western Pacific site for both mobile scenarios. Section V concludes.

II. Preliminaries

This section states the mobile-UWSN architecture and the acoustic, multipath, noise, and Q-learning objects that E-LQTP inherits from LQARP [9]. The inherited objects are restated so that the dual-track estimator of Section III can be written against a closed system model; no new channel or learning contribution is claimed. Subsection II-F isolates the estimation problem that mobility adds on top of that foundation.

A. Network Architecture

The network is a single-sink acoustic mesh of N autonomous underwater vehicles (AUVs) and one unmanned surface vehicle (USV) sink, deployed in a $L \times L \times H = 6\,\mathrm{km} \times 6\,\mathrm{km} \times 500\,\mathrm{m}$ Western Pacific water column (Fig. 1). The sink sits at the horizontal centroid with kinematic depth $z_{\mathrm{sink}} = 5\,\mathrm{m}$, so that it is uniquely shallowest for the depth-progressive candidate filter of Section II-E. Its acoustic query depth is independent of that kinematic depth and is set to the Bellhop source-depth slice $z_{\mathrm{ac}} = 50\,\mathrm{m}$ (a towed or hull-offset transducer). Each AUV s_i has kinematic state $(p_i, \dot{p}_i) \in \mathbb{R}^3 \times \mathbb{R}^3$, residual energy E_{s_i}, and a feasible-neighbor count number_{s_i}. Relays are placed uniformly in $[0, L]^2 \times [10, H]$ with a $1\,\mathrm{m}$ three-dimensional separation floor. The sink does not generate data. Every AUV emits a data packet of $l = 400\,\mathrm{bit}$ every $30\,\mathrm{s}$.

Two mobility fields are used in evaluation (Fig. 1) and are drawn at $N = 40$ in Fig. 4. Scenario I is the stationary identity: $\dot{p}_i \equiv 0$. Scenario II is a shared horizontal current $\dot{p}_i(t) = v_{\mathrm{cmd},i} + v_{\mathrm{drift}}$ with $v_{\mathrm{cmd},i} = 0$ and $\dot{p}_{\mathrm{sink}} = 0$, so that AUV–AUV relative velocity is identically zero and the only persistent range-rate is AUV-to-sink. Scenario III replaces the uniform current with a counterclockwise Rankine eddy centered on the sink plus an independent horizontal Gauss–Markov fluctuation, so that AUV–AUV v_{rel} is nonzero. A relay that leaves the valid horizontal envelope is removed; in-flight packets at that node fail as envelope-exit losses, not voids. Vertical current is not modelled: a $1\,\mathrm{m/s}$ vertical component would empty the $500\,\mathrm{m}$ column in a few hundred seconds.

B. Underwater Acoustic Channel

Instantaneous link quality is the sonar-equation SNR on a precomputed Bellhop field, inherited from LQARP [9]. For a transmitter–receiver pair (s_i, s_j),

$$\mathrm{SNR}(s_i, s_j) = \mathrm{SL} - \mathrm{TL}(r_H, z_q) - \mathrm{NL} + \mathrm{DI}, \tag{1}$$

where $\mathrm{SL} = 175\,\mathrm{dB}$ re $1\,\mu\mathrm{Pa@1\,m}$ is the source level, $\mathrm{DI} = 0\,\mathrm{dB}$ is the omnidirectional directivity index, and NL is the band-integrated Wenz level of Section II-D. Horizontal range $r_H = \|(x_i - x_j, y_i - y_j)\|$ and query depth $z_q = \max(z_j^{\mathrm{ac}}, 5\,\mathrm{m})$ index a January World Ocean Atlas 2023 / GEBCO cell at $20.125°\mathrm{N}$, $124.125°\mathrm{E}$ (seafloor $5524\,\mathrm{m}$, source depth $50\,\mathrm{m}$, carrier $24\,\mathrm{kHz}$, integration band $B = 2\,\mathrm{kHz}$). Transmission loss is the sonar-equation column of that grid,

$$\mathrm{TL}(r_H, z_q) = -10 \log_{10} \bar{G}(r_H, z_q), \tag{2}$$

with $\bar{G}$ the in-band power average of the equivalent transfer function reconstructed from Bellhop eigenrays (geometric spreading with $\kappa = 2$ and Thorp absorption as in LQARP [9]; Bellhop amplitudes are not reused, so spreading and absorption are not double-counted). A lookup that is non-finite is treated as a failed link: $\mathrm{SNR} \geq \mathrm{SNR}_{\mathrm{th}} = 8.68\,\mathrm{dB}$ is reception, and any non-finite or sub-threshold value is a failed link. The January $50\,\mathrm{m}$ crossing range at that threshold is $r_{\mathrm{range}} = 5125\,\mathrm{m}$. Every protocol variant—LQARP, E-LQTP, the RLOR-inspired baseline, and the Fixed-Periodic ablation—uses the same Bellhop lookup for every SNR decision [9].

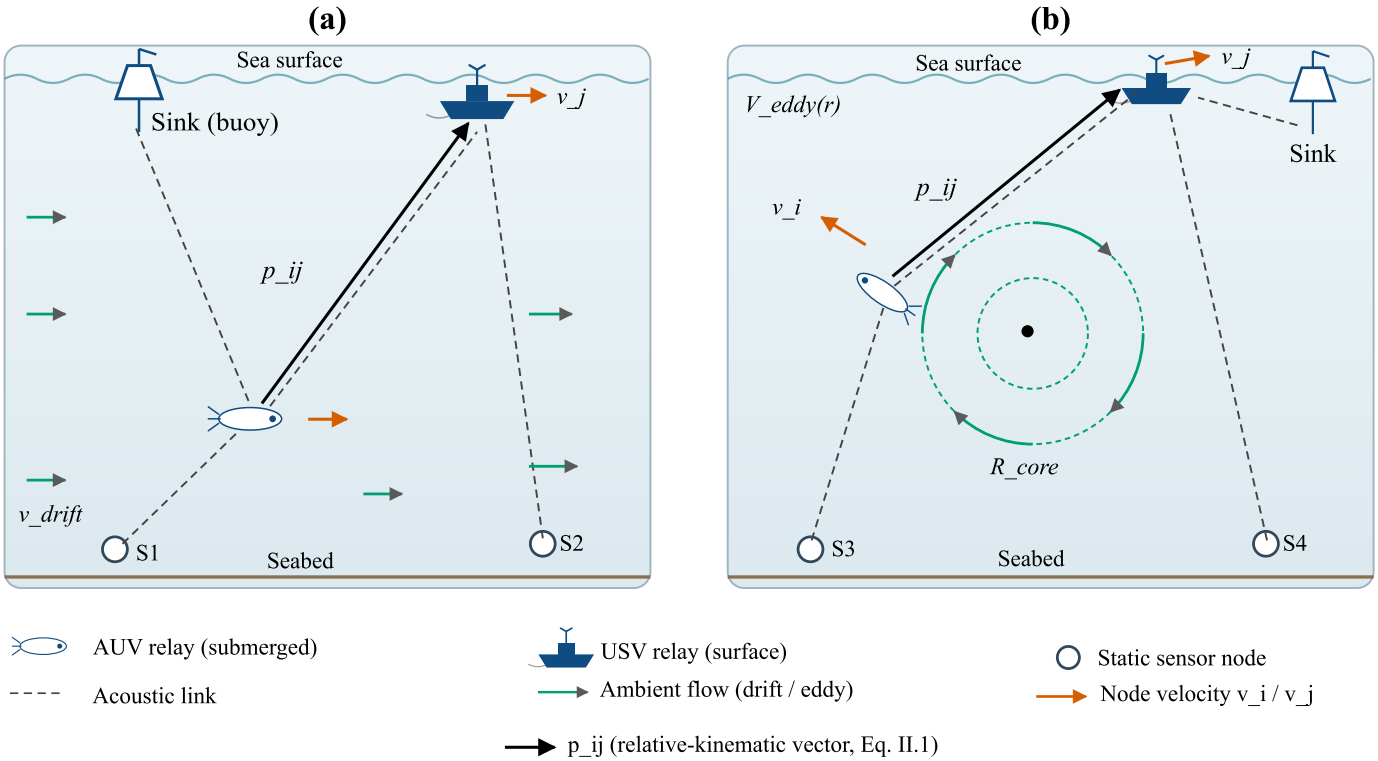


Fig. 1. Mobile UWSN. (a) Shared current. (b) Rankine eddy. Solid: relative kinematic vector p_{ij}. Dashed: acoustic link.

C. Multipath Representation

The channel is frequency-selective on this site. Water-column delay spread of about 3 ms at 1 km sets a coherence bandwidth of a few hundred hertz, much narrower than $B = 2\,\mathrm{kHz}$. The scalar used by routing is therefore the 2 kHz power average $\bar{G}$, not a single-frequency coherent snapshot. That average is the LQARP-style link budget; it does not resolve fast fading, and the median pre-filter of Section III-B is not claimed to do so either. Out-of-envelope (range, depth) cells are stored as non-finite and are rejected by the reception gate rather than interpolated.

D. Ambient Noise

Ambient noise follows the four-component Wenz model inherited from LQARP [9] at shipping factor $s = 0.2$ and wind $w = 2.0\,\mathrm{m/s}$ (assumed, not taken from WOA/GEBCO). The band-integrated level over 23–25 kHz is $\mathrm{NL} = 65.77\,\mathrm{dB}$ re $1\,\mu\mathrm{Pa}$. At 24 kHz the spectral densities (dB re $1\,\mu\mathrm{Pa}^2/\mathrm{Hz}$) are turbulence -24.4, shipping -13.0, wave 32.7, and thermal 12.6; wind and thermal terms dominate. NL is constant on the grid; motion enters (1) only through $\mathrm{TL}(r_H, z_q)$.

E. Q-learning Routing

LQARP treats each node s_i as a Q-learning agent [9] on the MDP (S, A, P, R). The action a_j selects neighbor s_j as next hop. A neighbor is a candidate only if it lies in all three of

$$\begin{aligned} N_{\mathrm{range}}(s_i) &= \{s_j : d(s_i, s_j) < r_{\mathrm{range}}\}, \\ N_{\mathrm{SNR}}(s_i) &= \{s_j : \mathrm{SNR}(s_i, s_j) > \mathrm{SNR}_{\mathrm{th}},\ j \neq i\}, \\ N_{\mathrm{depth}}(s_i) &= \{s_j : \mathrm{depth}_{s_i} - \mathrm{depth}_{s_j} > 0,\ j \neq i\}, \end{aligned} \tag{3}$$

so that $I(s_i) = N_{\mathrm{range}}(s_i) \cap N_{\mathrm{SNR}}(s_i) \cap N_{\mathrm{depth}}(s_i)$ and $A = I(s_i)$. The reward on a feasible action is

$$R^{a_j}_{s_i,s_j} = \frac{c_{\mathrm{ene}}(s_j)\, c_{\mathrm{del}}(s_i, s_j)\, c_{\mathrm{nei}}(s_j)}{c_{\mathrm{lin}}(s_i, s_j)}, \tag{4}$$

with

$$\begin{aligned} c_{\mathrm{lin}}(s_i, s_j) &= \ln\!\big(1/P_{\mathrm{succ}}(s_i, s_j)\big), \\ P_{\mathrm{succ}} &= (1 - p_e)^l, \\ c_{\mathrm{ene}}(s_j) &= (2/\pi)\arctan\!\big((E_{s_j} - E_{\mathrm{avg}})/E_{\mathrm{ini}}\big), \\ c_{\mathrm{del}}(s_i, s_j) &= (E_{s_j}/E_{\mathrm{ini}})\,(\mathrm{depth}_{s_i} - \mathrm{depth}_{s_j}), \\ c_{\mathrm{nei}}(s_j) &= \exp(\mathrm{number}_{s_j}). \end{aligned} \tag{5}$$

p_e is the $M = 2$ (BPSK) OFDM bit-error expression of LQARP Eq. 34, floored at $p_e = \max(p_e^{\mathrm{AWGN}}, 10^{-9})$ so that c_{lin} remains finite on the high-SNR plateau that is typical at short range on this grid ($c_{\mathrm{lin}} \approx 4 \times 10^{-7}$ for $l = 400$). E_{avg} is the mean residual energy over $I(s_i)$, not the global mean; $\mathrm{number}_{s_j} = |I(s_j)|$, and $c_{\mathrm{nei}} = 1$ when s_j has no onward candidate. The temporal-difference update is

$$Q(s_i, a_j) \leftarrow (1-\alpha)Q(s_i, a_j) + \alpha\Big[R^{a_j}_{s_i,s_j} + \gamma \max_{a \in I(s_j)} Q(s_j, a)\Big], \tag{6}$$

with $\alpha = 0.9$, $\gamma = 0.8$, and $s_i.\mathrm{NextHop} = \arg\max_{a_j \in I(s_i)} Q(s_i, a_j)$. Q-values start at zero.

When $I(s_i) = \emptyset$ at a non-sink node, LQARP (and E-LQTP) compute a relaxed detour set $D(s_i) = N_{\mathrm{range}}(s_i) \cap N_{\mathrm{SNR}}(s_i)$ and keep only two-hop-viable, unvisited members, with a cap of three detour hops per packet. That mechanism is a bounded reading of LQARP's qualitative two-hop remark, applied identically to LQARP, E-LQTP, and Fixed-Periodic; it is not a new contribution.

F. Relative-Motion Estimation

LQARP evaluates $I(s_i)$ and the four reward factors from neighbor state. Under mobility a deployed node must decide when to re-advertise velocity and residual energy, because a periodic HELLO spends airtime while the acoustic link is still accurate and a silent node would leave that advertisement unchanged. In the campaign reported here, no neighbor table ages. At every forwarding decision and every EKF tick, LQARP, E-LQTP, the RLOR-inspired baseline, and Fixed-Periodic read the simulator's current pose together with a fresh Bellhop sample. That sample is the ground-truth channel query used for reception, available at every simulation tick $dt = 1\,\mathrm{s}$. The HELLO of Section III-B is the charged velocity copy that resets the forecast. $I(s_i)$ is taken from the live pose. Ordinary event-triggered Kalman filtering [15]–[18], [41] tests an innovation $z_k - h(\hat{x}_{k|k-1})$ and therefore *cannot* accumulate staleness here: the measurement-corrected track heals the only observable dimension (horizontal range) on every tick, whether or not a HELLO has ever been heard. The one-step Kalman innovation e_k is consequently a noise residual, not a ZOH-velocity residual, for any finite threshold. That is why E-LQTP keeps a second forecast track, propagated without a measurement update, and triggers on the forecast-versus-truth innovation of Section III-B, following the predicted-information idea of Li *et al.* [15] but scheduling a routing HELLO rather than a tracking packet. The $2\,\mathrm{dB}$ cap on that trigger threshold is a choice of this paper. It lies below the LOON21 Kalman mean absolute SNR errors of Busacca *et al.*, which are 4.81, 3.47, and $3.24\,\mathrm{dB}$ on the three tests cited in Section I [19]. The filter is a staleness detector for HELLO, not an SNR-RMSE competitor.

III. E-LQTP Design

E-LQTP is LQARP plus a dual-track relative-motion estimator, a persistence-gated HELLO, an optional pair of reward modifiers, and a stability pre-filter on the candidate set. The validated operating point uses the estimator and the trigger and sets the reward modifiers to zero. Fig. 2 is the per-tick mechanism; Algorithm 1 is the same loop in the equation-referenced form of LQARP's Algorithm 1 [9].

A. Kinematic State

For each directed pair (s_i, s_j),

$$\begin{aligned} p_{ij}(t) &= p_i(t) - p_j(t), \\ \dot{p}_{ij}(t) &= \dot{p}_i(t) - \dot{p}_j(t), \\ v_{\mathrm{rel}}(i,j) &= \big(\dot{p}_{ij} \cdot p_{ij}\big)/\|p_{ij}\|. \end{aligned} \tag{7}$$

Negative v_{rel} is closing range. Own velocity $\dot{p}_i$ is odometry and is known exactly. Neighbor velocity $\dot{p}_j$ is a zero-order hold. Before any hold has been written the hold is $(0,0,0)$. Under the shared current of Scenario II that initialization assigns pair (i,j) the relative velocity $\dot{p}_i$, which is not the true relative velocity of two relays drifting together. The assignment is intentional: writing j's true velocity at $t = 0$ would make every AUV–AUV pair a perfect match to the constant-velocity model and would leave the trigger with no innovation on those pairs. On every later tick for which $\mathrm{Broadcast}_k(i,j) = 1$, including ticks on which the flag stays high after the rising edge, the three velocity components of the corrected track are overwritten with no blending, $\hat{x}^{(v)}_{k|k} \leftarrow \dot{p}_i - \dot{p}_j^{\mathrm{live}}$, and the forecast track is reset to that corrected state. The optional drift term does not use the initial hold. Until the first overwrite, β_2 is omitted and $c_{\mathrm{drift}} = c_{\mathrm{del}}$. SNR does not observe the velocity block. Horizontal range is adjusted only through the predict Jacobian of Section III-B. The velocity hold is therefore the sole source of v_{rel} for the optional drift term.

Scenario II kinematics are $\dot{p}_i(t) = v^{(0)}_{\mathrm{drift}}$ with $\|v^{(0)}_{\mathrm{drift}}\|$ swept in $[0.5, 3.5]\,\mathrm{m/s}$ (reported midpoint $2\,\mathrm{m/s}$) and one horizontal azimuth, drawn once per run and shared by every relay. Scenario III uses a Rankine-capped counterclockwise eddy of core radius $R_{\mathrm{core}} = 2000\,\mathrm{m}$ and speed $V_{\mathrm{eddy}} = 2\,\mathrm{m/s}$ centered at $(3000, 3000)\,\mathrm{m}$,

$$v_\theta(r) = V_{\mathrm{eddy}} \min(r/R_{\mathrm{core}}, 1), \tag{8}$$

plus the discrete Gauss–Markov step

$$v^i_{k+1} = \zeta v^i_k + (1-\zeta)\, v_{\mathrm{eddy}}(p_i) + \sqrt{1-\zeta^2}\,\sigma \odot w^i_k, \tag{9}$$

with $\zeta = 0.95$, $dt = 1\,\mathrm{s}$, $\sigma = (1,1,0)\,\mathrm{m/s}$, and $w^i \sim \mathcal{N}(0, I)$. The zero vertical innovation is a modelling choice: mid-latitude mesoscale flow at these scales is geostrophic, and a $1\,\mathrm{m/s}$ vertical term would empty the column. Inside the solid-body core the mean-flow pair difference is perpendicular to the line of sight; the Gauss–Markov innovation supplies most of the observable radial projection (mean $|v_{\mathrm{radial}}| \approx 1.06\,\mathrm{m/s}$ against mean $|v_{\mathrm{tangential}}| \approx 2.54\,\mathrm{m/s}$ on the kinematics-only diagnostic).

B. Dual-Track EKF and Event Trigger

Each neighbor pair carries a seven-dimensional state

$$x_k = \big[p_{ij}^\top,\, \dot{p}_{ij}^\top,\, \mathrm{SNR}_{ij}\big]^\top. \tag{10}$$

The position–velocity block is constant-velocity,

$$F_{\mathrm{kin}} = \begin{bmatrix} I_3 & dt\, I_3 \\ 0_3 & I_3 \end{bmatrix}, \tag{11}$$

and does not carry SNR. Both tracks apply the same nonlinear predict. Writing $p^+ = p + dt\,\dot{p}$ and $r_H^+ = \|(p_x^+, p_y^+)\|$ for the predicted horizontal range, with z_q the neighbor's acoustic query depth,

$$\begin{aligned} p^+ &= p + dt\,\dot{p}, & \dot{p}^+ &= \dot{p}, \\ \mathrm{SNR}^+ &= \begin{cases} \mathrm{SNR}(r_H^+, z_q), & \text{if the lookup is finite,} \\ \mathrm{SNR}^-, & \text{otherwise,} \end{cases} \end{aligned} \tag{12}$$

The lookup is (1). An out-of-envelope cell keeps the last finite SNR; it is not interpolated. The covariance is propagated by the 7×7 Jacobian F of (12): $F = I_7$, the three constant-velocity couplings are $F_{1,4} = F_{2,5} = F_{3,6} = dt$, the SNR row is replaced rather than scaled so $F_{7,7} = 0$, and the only nonzero SNR–position entries are the horizontal radial derivatives

$$F_{7,1} = \frac{\partial \mathrm{SNR}}{\partial r_H}\frac{p_x^+}{r_H^+}, \qquad F_{7,2} = \frac{\partial \mathrm{SNR}}{\partial r_H}\frac{p_y^+}{r_H^+}.$$

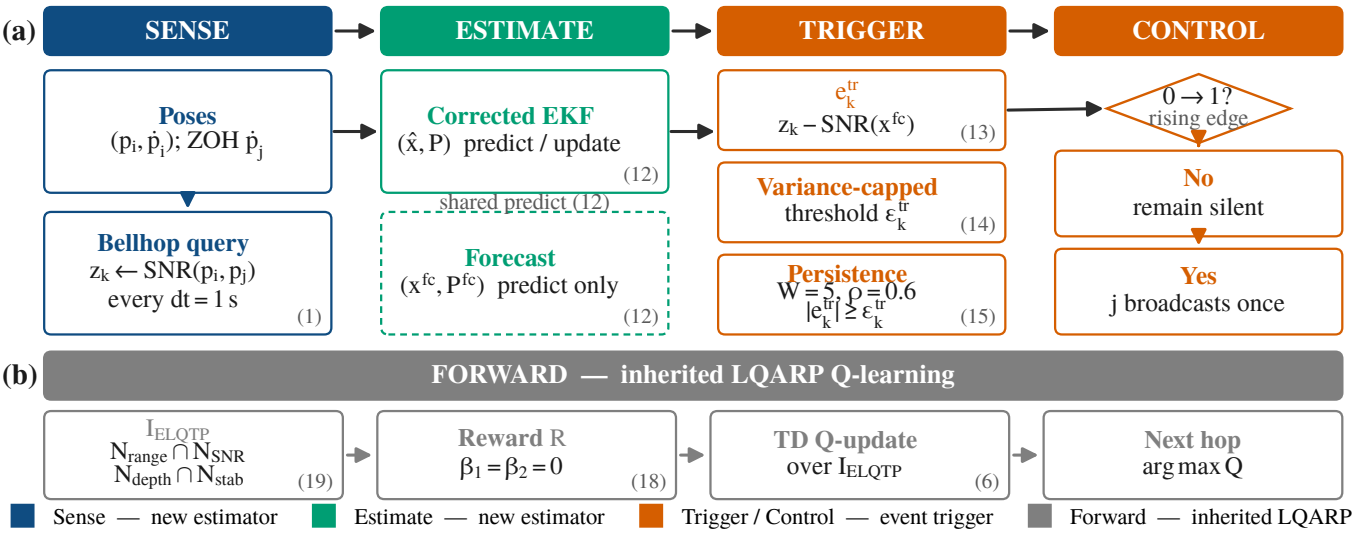


Fig. 2. E-LQTP per-tick mechanism. (a) Discovery: sensing, dual-track estimation, event trigger, and rising-edge HELLO. (b) Forwarding at $\beta_1 = \beta_2 = 0$, inherited from LQARP.

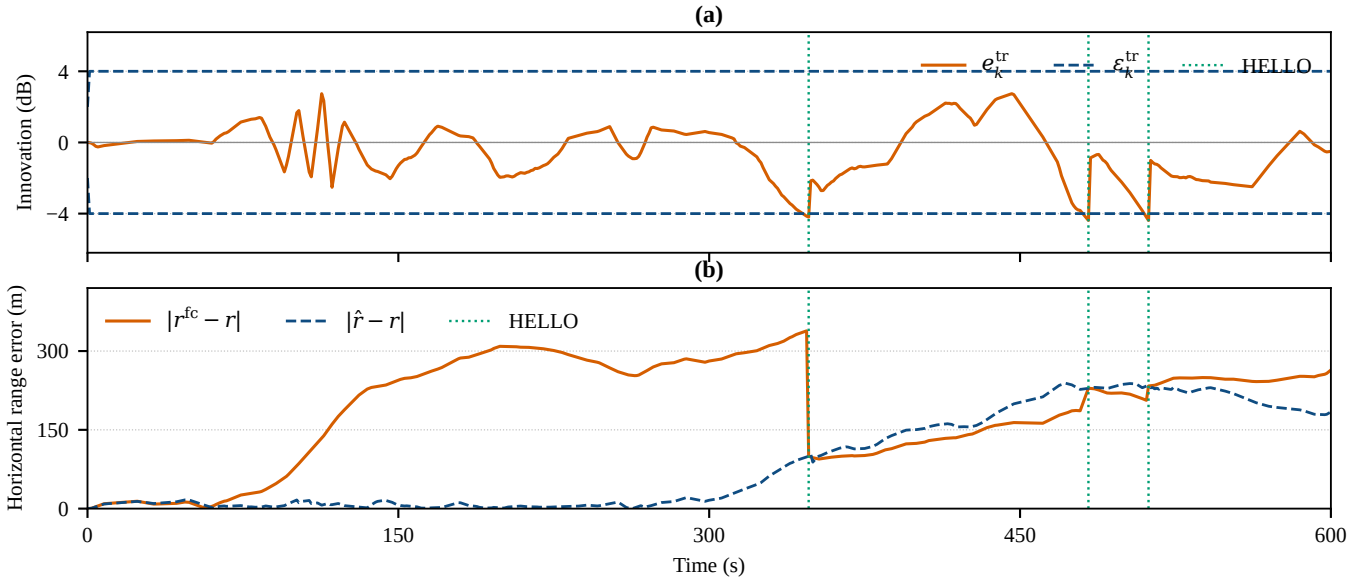


Fig. 3. Dual-track trigger, pair $(1, 2)$. (a) e_k^{tr} and $\pm\varepsilon_k^{\mathrm{tr}}$. (b) Range error. HELLOs at 348, 483, 512 s.

The range derivative is a central difference with step 10 m. Process noise is then added on the diagonal, $P^+ = FPF^\top + Q$, with $q_{\mathrm{pos}} = 1.0$, $q_{\mathrm{vel}} = 0.1$, and $q_{\mathrm{SNR}} = 1.0$. The measurement variance is $R = r_{\mathrm{meas}} = 4.0\,\mathrm{dB}^2$.

Two tracks run in parallel and share (12). The *corrected* track $(\hat{x}, P)$ also applies a measurement update. The raw sample z_k is the Bellhop SNR at the nodes' true current positions, computed every dt whether or not a packet is heard. The median $\tilde{z}_k$ is taken over the samples stored for that pair, at most $W_{\mathrm{mp}} = 3$; the first ticks use the shorter prefix. The one-step innovation $e_k = \tilde{z}_k - \hat{x}^{\mathrm{SNR}}_{k|k-1}$ enters the filter and, optionally, c_{pred}. The update uses $H = [0\,0\,0\,0\,0\,0\,1]$ because the measurement is the SNR component. The gain is the corresponding column of P, $K = P_{:,7}/S_k$ with $S_k = P_{7,7} + r_{\mathrm{meas}}$, so the innovation moves every state component that has covariance with SNR. Horizontal range is among those components because $F_{7,1}$ and $F_{7,2}$ couple predicted range into the SNR variance. The tangential directions and the velocity block are not directly observed by H. The *forecast* track $(x^{\mathrm{fc}}, P^{\mathrm{fc}})$ receives the predict and no measurement update. It is reset to $(\hat{x}_{k|k}, P_{k|k})$ on every tick with $\mathrm{Broadcast}_k(i, j) = 1$, after the corrected-track update, by the velocity overwrite of Section III-A. The trigger innovation uses the raw query against the forecast SNR, which (12) has just set to the lookup at the forecast range,

$$e_k^{\mathrm{tr}} = z_k - \mathrm{SNR}\big(r_H(x_k^{\mathrm{fc}}), z_q\big). \tag{13}$$

A one-step e_k cannot form this residual: the corrected track is healed every tick, whereas x^{fc} accumulates the motion implied by the held velocity.

The corrected-track threshold is left uncapped. The forecast covariance is not measurement-updated, so $\varepsilon_0+\lambda\sqrt{S_k^{\mathrm{tr}}}$ would make a long-silent pair progressively harder to trigger. The trigger threshold therefore caps the uncertainty term,

$$\begin{aligned}\varepsilon_k &= \varepsilon_0 + \lambda\sqrt{S_k},\\ \varepsilon_k^{\mathrm{tr}} &= \varepsilon_0 + \lambda\min\bigl(\sqrt{S_k^{\mathrm{tr}}}, s_{\mathrm{cap}}\bigr),\end{aligned} \tag{14}$$

with $\varepsilon_0 = 2.0\,\mathrm{dB}$, $\lambda = 1$, $s_{\mathrm{cap}} = \sqrt{r_{\mathrm{meas}}} = 2.0\,\mathrm{dB}$, $S_k = P_{7,7} + r_{\mathrm{meas}}$, and $S_k^{\mathrm{tr}} = P_{7,7}^{\mathrm{fc}} + r_{\mathrm{meas}}$. The cap is the design choice identified in Section I; it is not taken from [19]. A persistence gate with width $W = 5$ and fraction $\rho = 0.6$ then sets the flag. Let n_k be the number of finite trigger samples retained for the pair, at most W. Until five samples have been stored, n_k is that shorter prefix, so a single early exceedance has fraction one and can pass ρ.

$$\begin{aligned}\mathrm{Broadcast}_k = 1 \quad &\text{if } |e_k^{\mathrm{tr}}| \ge \varepsilon_k^{\mathrm{tr}}\\ &\text{and } \tfrac{1}{n_k}\textstyle\sum_{m\in\mathcal{W}_k}\mathbf{1}\{|e_m^{\mathrm{tr}}| \ge \varepsilon_m^{\mathrm{tr}}\} \ge \rho,\\ \mathrm{Broadcast}_k = 0 \quad &\text{otherwise,}\end{aligned} \tag{15}$$

where $\mathcal{W}_k$ is the retained window. A non-finite z_k skips the update and forces $\mathrm{Broadcast}_k = 0$. Node j originates one HELLO on a tick if $\mathrm{Broadcast}_k(i,j)$ rises for any observer i. Ticks on which the flag merely stays high do not add a further transmission. Those ticks do repeat the velocity overwrite and the forecast reset.

Fig. 3 shows the two tracks on Scenario III pair $(1,2)$. Panel (a) is e_k^{tr} against $\pm\varepsilon_k^{\mathrm{tr}}$; the threshold sits at the s_{cap} ceiling once forecast variance has grown. Rising-edge HELLOs occur at $t = 348$, 483, and 512 s. Panel (b) is the corresponding horizontal range error. The forecast range error grows while the flag is low and is reset when the flag is high. The corrected range error stays smaller because the per-tick query, acting through K, heals the radial state. That is the mechanism the control-plane results of Section IV attribute the overhead reduction to.

C. Reward Terms

E-LQTP replaces c_{lin} and c_{del} with

$$c_{\mathrm{pred}} = c_{\mathrm{lin}} + \beta_1\max\bigl(0, e_k - e_{k-1}\bigr), \tag{16}$$

$$c_{\mathrm{drift}} = c_{\mathrm{del}}\bigl(1 + \beta_2\,\sigma(-v_{\mathrm{rel}})\bigr), \tag{17}$$

where $\sigma(u) = (1+e^{-u})^{-1}$. If i has not yet heard j, the β_2 factor is omitted ($c_{\mathrm{drift}} = c_{\mathrm{del}}$). The composite reward is

$$R_{s_i,s_j}^{a_j} = \frac{c_{\mathrm{ene}}(s_j)\,c_{\mathrm{drift}}(s_i,s_j)\,c_{\mathrm{nei}}(s_j)}{c_{\mathrm{pred}}(s_i,s_j)}. \tag{18}$$

$\beta_1 = \beta_2 = 0$ recovers (4) exactly and is the validated default. The Q-update is (6) over the restricted action set of Section III-D.

The additive form (16) is scale-inconsistent on the BER-floor plateau: a typical one-step Δe_k is $\mathcal{O}(0.01)\,\mathrm{dB}$, so an $\mathcal{O}(1)$ β_1 overrides $c_{\mathrm{lin}} \approx 4\times10^{-7}$ by four orders of magnitude. A blend-scale β_1 (10^{-5}–10^{-6}) has no favorable effect either (Section IV-F). The one-step Kalman trend persists for only about one second (median same-sign run 1–4 s, 25th percentile 1 s), comparable to or shorter than a hop (≈ 2 s). c_{drift} has no scale bug, but under the eddy the radial-velocity sign persists for a median 3 s (25th percentile 1 s), so the persistent-closing-range assumption does not hold. Both findings are scoped negative results.

Algorithm 1 E-LQTP: Event-Triggered HELLO and Q-Learning Forwarding

Require: poses $(p_i, \dot{p}_i)$; hold $\dot{p}_j \leftarrow 0$ until written; $(\hat{x}, P)$, $(x^{\mathrm{fc}}, P^{\mathrm{fc}})$
Ensure: one HELLO per rising node; $s_i.\mathrm{NextHop}$
Discovery: dual-track predict (12)
for each directed pair (i,j) **do**
  Predict both tracks by (12), using (11)
  $z_k \leftarrow \mathrm{SNR}(p_i,p_j)$ (1) // [9]
  if z_k is non-finite **then**
    skip update; $\mathrm{Broadcast}_k(i,j) \leftarrow 0$
  else
    $\tilde{z}_k \leftarrow$ median of the stored prefix, at most W_{mp}
    $e_k \leftarrow \tilde{z}_k - \hat{x}^{\mathrm{SNR}}$; e_k^{tr} from (13)
    ε_k and $\varepsilon_k^{\mathrm{tr}}$ from (14)
    EKF-update $(\hat{x}, P)$ on $\tilde{z}_k$ with $H = [0\ \cdots\ 1]$
    $\mathrm{Broadcast}_k(i,j)$ from (15)
    if $\mathrm{Broadcast}_k(i,j) = 1$ **then**
      $\hat{x}^{(v)} \leftarrow \dot{p}_i - \dot{p}_j^{\mathrm{live}}$; reset forecast to $(\hat{x}, P)$
    end if
  end if
end for
if event-triggered **then**
  node j sends one HELLO if any i has a $0 \to 1$ edge on (i,j)
else if Fixed-Periodic **and** $t \bmod T_{\mathrm{ctrl}} = 0$ **then**
  every node sends one HELLO; every pair ingests $\dot{p}_j^{\mathrm{live}}$
end if
Forwarding: LQARP Q-learning [9]
if node s_i has a data packet **then**
  I_{ELQTP} from (3), (19); detour if empty // [9]
  reward (18); TD update (6) // [11]
  $s_i.\mathrm{NextHop} \leftarrow \arg\max Q$
end if

D. Candidate Set

E-LQTP adds a fourth filter to (3),

$$\begin{aligned}N_{\mathrm{stab}}(s_i) = &\bigl\{s_j : \mathrm{Broadcast}_k(i,j) = 0\bigr\}\\ &\cup \bigl\{s_j : |e_k(i,j)| < \varepsilon_k(i,j)\bigr\},\end{aligned} \tag{19}$$

and forwards on $I_{\mathrm{ELQTP}}(s_i) = N_{\mathrm{range}} \cap N_{\mathrm{SNR}} \cap N_{\mathrm{depth}} \cap N_{\mathrm{stab}}$. Here e_k and ε_k are the corrected-track innovation and the uncapped threshold in (14), not $(e_k^{\mathrm{tr}}, \varepsilon_k^{\mathrm{tr}})$. The leftover 0.2% hop difference versus LQARP on Scenario III is this pre-filter with the trigger on, not a reward term. The detour of Section II-E is unchanged.

E. Packets and Control Accounting

A data packet is $l = 400\,\mathrm{bit}$ at $R_b = 1000\,\mathrm{bit/s}$ (airtime 0.4 s) and carries the payload plus a visited-node list for detour loop prevention (TTL 32). A dedicated HELLO is charged at the same 400 bit. Its timing, the single transmission by node

j, and the oracle velocity copy are those of Section III-B: the payload is not demodulated, and the copy is written only into the velocity hold. Forwarding for every protocol reads the simulator's current pose, residual energy, and a fresh Bellhop sample at the decision instant, as stated in Section II-F. LQARP and the RLOR-inspired baseline generate no dedicated control frames, so their O_{ctrl} is identically zero. E-LQTP and Fixed-Periodic add HELLO bytes on top of that shared data plane. HELLO joules enter O_{ctrl} only. A control transmission leaves residual energy and data-packet reception unchanged. Control overhead is

$$O_{\mathrm{ctrl}} = \frac{\text{control bytes on } [0, T_{\mathrm{sim}}]}{T_{\mathrm{sim}}}. \tag{20}$$

Drain-period control traffic is not billed. Fixed-Periodic replaces (15) with the grid $t = 0, T_{\mathrm{ctrl}}, 2T_{\mathrm{ctrl}}, \ldots$ at $T_{\mathrm{ctrl}} = 5\,\mathrm{s}$ and is otherwise identical, so the ablation isolates timing.

F. Simulation Implementation

The protocols are implemented in ns-3.41 with Aqua-Sim-NG and share one Bellhop lookup path [9]. Scenario I holds every node fixed; Scenarios II and III use constant-velocity kinematics with poses read at every forwarding event and every EKF tick. The same random placement is used for all four protocols. Reception is an SNR-threshold test on the shared Bellhop sample. Collisions, backoff, and idle listening are omitted, so control airtime does not delay or drop a data packet. The RLOR-inspired baseline follows the brief sketch in [9] and the opportunistic class surveyed in [4]. That sketch does not specify a line-by-line algorithm, so the baseline used here takes the candidate set $N_{\mathrm{range}} \cap N_{\mathrm{SNR}}$ (no N_{depth}, no detour) and forwards $\arg\max P_{\mathrm{succ}}$ with no Q-table.

IV. Performance Evaluation

This section reports the Western Pacific campaign that is the empirical claim of the paper. Every reported value is a direct statistic over the recorded simulation runs described in Table I; no cell is interpolated or invented.

A. Simulation Configuration

All four protocols share the constants of Table I: $N = 40$ relays in the $6\,\mathrm{km} \times 6\,\mathrm{km} \times 500\,\mathrm{m}$ column, January WOA23/GEBCO/Bellhop cell of Section II-B, $\mathrm{SL} = 175\,\mathrm{dB}$, $R_b = 1000\,\mathrm{bit/s}$, $l = 400\,\mathrm{bit}$, $M = 2$, $\mathrm{SNR_{th}} = 8.68\,\mathrm{dB}$, $r_{\mathrm{range}} = 5125\,\mathrm{m}$, $E_{\mathrm{ini}} = 10000\,\mathrm{J}$, $\alpha = 0.9$, $\gamma = 0.8$, and SNR-threshold reception with no modeled contention. A node that leaves the area is removed. Mobile runs use $T_{\mathrm{sim}} = 600\,\mathrm{s}$ and five independent random seeds. The validated E-LQTP point is the event trigger on and $\beta_1 = \beta_2 = 0$. Fixed-Periodic uses the same reward weights and $T_{\mathrm{ctrl}} = 5\,\mathrm{s}$. That interval is one end-to-end delay of the $v = 2\,\mathrm{m/s}$ mesh ($T_{\mathrm{e2e}} \approx 4.4\,\mathrm{s}$; one-hop flight $\approx 2.0\,\mathrm{s}$ on the Scenario III β sweep). It is a single, drift-independent budget against which an event-triggered scheme can be cheaper or more expensive; it is not claimed to be the optimal periodic timer, and it is the same class of timer used to rebuild STGR's spatial-temporal graph [10] and to refresh QTAR and SAQR neighbor state [11], [12].

TABLE I
Shared simulation parameters. E-LQTP-only knobs are marked.

Parameter	Value
Volume / N / seeds	$6\,\mathrm{km} \times 6\,\mathrm{km} \times 500\,\mathrm{m}$ / 40 / five
Site / season	20.125°N, 124.125°E, January
Seafloor / carrier / band	5524 m / 24 kHz / 2 kHz
SL / DI / $\mathrm{SNR_{th}}$	175 dB / 0 dB / 8.68 dB
R_b / l / M	1000 bit/s / 400 bit / 2
r_{range} / E_{ini}	5125 m / 10000 J
Efficiency / P_{rx} / BER floor	0.5 / 0.1 W / 10^{-9}
Data interval / T_{sim} / dt	30 s / 600 s / 1 s
α / γ / reception / envelope	0.9 / 0.8 / SNR gate / leave-area nodes removed
ε_0 / s_{cap} / λ	2.0 dB / 2.0 dB / 1.0
W / W_{mp} / ρ / T_{ctrl} / (β_1, β_2)	5 / 3 / 0.6 / 5 s / (0, 0)

Fig. 4 is the velocity field of the two mobile scenarios on the same $N = 40$ placement and the same fixed sink. Panel (a) is shared current (Scenario II): every relay has the same horizontal velocity, so the true AUV–AUV range rate is identically zero. The filter does not start from that velocity. Section III-A initializes the neighbor-velocity hold at zero, and the hold takes the true value only after $\mathrm{Broadcast}_k = 1$. Panel (b) is the Rankine eddy (Scenario III): arrow length scales with local tangential speed, solid-body inside R_{core} and $1/r$ outside. The figure's $1/r$ decay is the classical Rankine sketch; the simulation itself uses the speed cap of (8).

Three scenarios are reported. Scenario I is the stationary identity of Section IV-C ($T_{\mathrm{sim}} = 1600\,\mathrm{s}$, EKF not ticked, trigger off) and is a reproduction check, not a primary mobility result. Scenario II sweeps $\|v_{\mathrm{drift}}\|$ at 0.5, 1.0, 2.0, 3.0, and $3.5\,\mathrm{m/s}$. Scenario III uses $V_{\mathrm{eddy}} = 2\,\mathrm{m/s}$ and $R_{\mathrm{core}} = 2000\,\mathrm{m}$.

B. Metrics

Packet-delivery ratio (PDR) is $N_{\mathrm{succ}}/N_{\mathrm{gen}}$ on packets whose source is still live. End-to-end delay T_{e2e} and hop count are means over successful deliveries. Energy per bit E_{bit} follows LQARP's Eq. 45 on *data* transmissions only [9]. Each data packet of $l = 400\,\mathrm{bit}$ at $R_b = 1000\,\mathrm{bit/s}$ has airtime $0.4\,\mathrm{s}$. Radiated acoustic power from $\mathrm{SL} = 175\,\mathrm{dB}$ re $1\,\mu\mathrm{Pa@1\,m}$ is $P_{\mathrm{ac}} = 10^{(\mathrm{SL}-170.8)/10} = 2.630\,\mathrm{W}$; with projector efficiency 0.5 the electrical transmit energy is $E_{\mathrm{tx}} = 2.104\,\mathrm{J}$ and the receive energy on a successful decode is $E_{\mathrm{rx}} = 0.040\,\mathrm{J}$. Idle listening is not modelled, so E_{bit} measures forwarding work, not modem duty cycle.

A dedicated HELLO uses the same $l = 400\,\mathrm{bit}$ as a data packet. LQARP and RLOR-inspired emit no dedicated control frames, so their O_{ctrl} is identically zero. Section III-E records that zero as an empty control-byte count on the shared live pose. E-LQTP and Fixed-Periodic add HELLO bytes on top of the same data plane, whose energy is already inside E_{bit}. Because HELLO length equals l, control transmit power is the linear image of O_{ctrl}:

$$P_{\mathrm{ctrl}} = \frac{O_{\mathrm{ctrl}}}{l/8}\, E_{\mathrm{tx}} = O_{\mathrm{ctrl}} \times 4.208 \times 10^{-2}\,\mathrm{W \cdot s/byte}. \tag{21}$$

At the reported operating points this is $5.69\,\mathrm{W}$ network-wide under Scenario II ($135.1\,\mathrm{byte/s}$) and $8.88\,\mathrm{W}$ under Scenario III ($211.0\,\mathrm{byte/s}$), against a drift-independent Fixed-Periodic budget of $17.40\,\mathrm{W}$ ($413.42\,\mathrm{byte/s}$, 41 transmitters $\times$ 121 grid instants on $[0, 600]\,\mathrm{s}$). The joule ratios are therefore

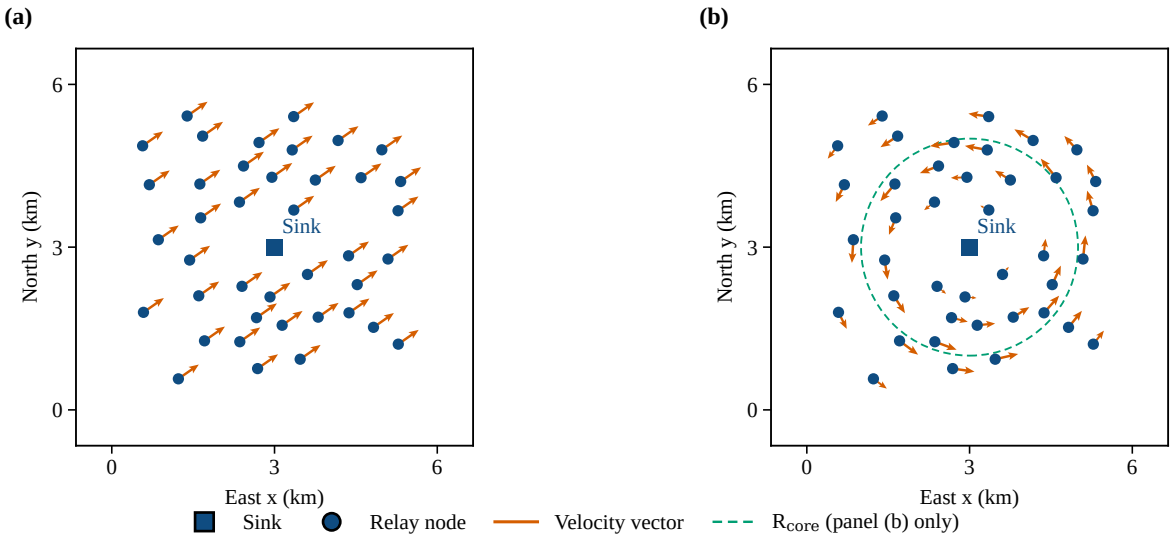


Fig. 4. Velocity fields, $N = 40$. (a) Uniform current, 2 m/s. (b) Rankine eddy, $R_{\mathrm{core}} = 2000$ m.

the same $0.33\times$ and $0.51\times$ already reported in bytes per second.

The ratio of means that adds the mean HELLO transmit energy, at $E_{\mathrm{tx}} = 2.104$ J, to the mean data-plane energy and divides by the mean number of delivered bits would raise Scenario II E_{bit} from 1.156×10^{-2} to 2.342×10^{-2} J/bit for E-LQTP and to 4.784×10^{-2} J/bit for Fixed-Periodic. LQARP would remain at 1.156×10^{-2} because it sends no dedicated control frames. We keep the LQARP definition of E_{bit} in the protocol-comparison tables so the data-plane columns remain comparable to the parent paper, and we report HELLO cost as O_{ctrl} (and the P_{ctrl} conversion above).

First-fire is the fraction of live AUV–AUV pairs that experience at least one rising-edge HELLO. VOID counts empty-candidate drops. Node-exit packet losses count packets whose forwarding node has left the Bellhop envelope; they are recorded separately from SNR-reception failures. n_{exit} counts relays that leave the valid envelope. Envelope attrition is a geometry fact, not a protocol difference. Under Scenario II the mean number of relays that exit by T_{sim} is 1.8, 4.4, 9.4, 14.4, and 16.0 of 40 at 0.5, 1.0, 2.0, 3.0, and 3.5 m/s. PDR in Table IV is therefore computed on a shrinking live set; LQARP and E-LQTP lose the same nodes.

C. Scenario I: Stationary Identity

Scenario I freezes every node ($v_{\mathrm{cmd}} = v_{\mathrm{drift}} = 0$) for $T_{\mathrm{sim}} = 1600$ s and $N_{\mathrm{gen}} = 2160$ at $N = 40$. E-LQTP is run in the identity configuration (trigger off, ε_0 huge, EKF not ticked) so that any departure from LQARP would be a code defect rather than a mobility effect [9]. Table II reports the five-seed means. LQARP and identity E-LQTP are bit-identical on every seed: PDR 1.000, hops 1.530 ± 0.068, delay 3.078 ± 0.071 s, $E_{\mathrm{bit}} = 8.20 \pm 0.37 \times 10^{-3}$ J/bit, $O_{\mathrm{ctrl}} = 0$, VOID $= 0$. Fixed-Periodic records 442.03 byte/s on this frozen field. A nominal budget of one 400-bit HELLO from each of 41 nodes every 5 s is 410 byte/s. The recorded stationary rate exceeds that budget. The mobile comparison in Section IV-B uses

TABLE II
SCENARIO I IDENTITY ($N = 40$, $T_{\mathrm{sim}} = 1600$ s, FIVE SEEDS). E-LQTP TRIGGER OFF; PDR $= 1.000$, VOID $= 0$.

Protocol	Hops	T_{e2e} (s)	E_{bit}	O_{ctrl}
LQARP	1.530 ± 0.068	3.078 ± 0.071	$8.20 \pm 0.37 \times 10^{-3}$	0
E-LQTP	1.530 ± 0.068	3.078 ± 0.071	$8.20 \pm 0.37 \times 10^{-3}$	0
RLOR-insp.	1.000 ± 0	1.887 ± 0.108	5.36×10^{-3}	0
Periodic	1.546 ± 0.039	3.119 ± 0.188	$8.29 \pm 0.21 \times 10^{-3}$	442.03

the exact grid 413.42 byte/s (41 transmitters $\times$ 121 instants on $[0, 600]$ s). A timer still spends that control traffic on a field that cannot change, the same class of waste a timer-rebuilt STG or a periodic QTAR HELLO incurs on a frozen mesh [10], [11]. RLOR-inspired always takes the one-hop sink path (hops $= 1.000$) because it drops N_{depth} and forwards $\arg\max P_{\mathrm{succ}}$; that is the opportunistic construction surveyed in [4], not a routing win claimed against LQARP. PDR saturates for all four protocols, which is the intended ceiling of a stationary, range-rich column, not a primary comparison.

D. Scenario II: Uniform Current Drift

Table III and Fig. 5 compare the four protocols at the reported midpoint $\|v_{\mathrm{drift}}\| = 2$ m/s. E-LQTP reproduces LQARP's data plane—PDR 0.9992 ± 0.0019, hops 2.154 versus 2.153, delay 4.439 s versus 4.438 s, identical E_{bit} at the printed precision, identical node-exit packet losses (0.60 ± 1.34), VOID $= 0$—and cuts control traffic from the analytic 413.42 byte/s grid to 135 ± 23 byte/s ($0.33\times$, never above 163). First-fire is 0.782 ± 0.042. By the definition in Section IV-B this is the fraction of live AUV–AUV pairs that record at least one rising edge. The true AUV–AUV range rate is zero once both velocities are known. Those pairs still fire because the velocity hold begins at zero, so the forecast initially integrates the observer's own drift. After the hold is written, the constant-velocity model matches the shared current. Links to the fixed sink retain a nonzero range rate after that write. Fixed-Periodic at $\beta_1 = \beta_2 = 0$ is LQARP's data plane plus that flat HELLO grid on the shared live pose of Section III-E, which isolates the timer from the reward.

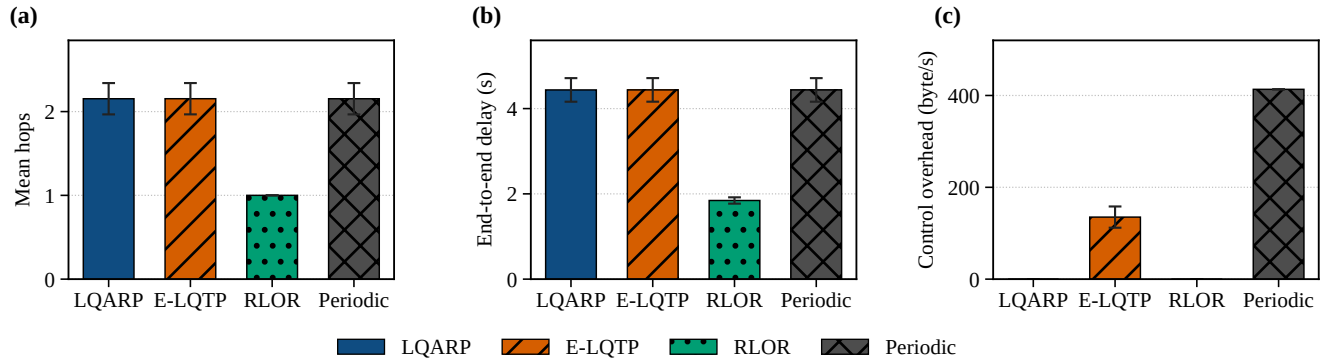


Fig. 5. Scenario II at $2\,\mathrm{m/s}$: hops, delay, and O_{ctrl}. Error bars are the five-seed sample standard deviation.

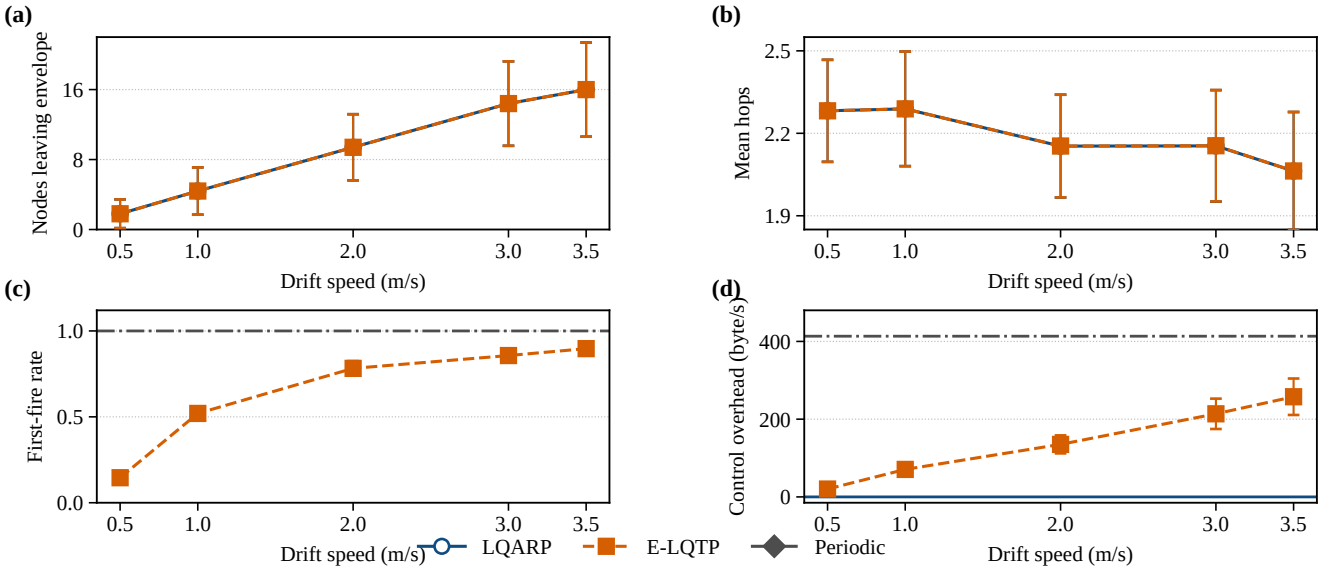


Fig. 6. Scenario II versus drift: exits, hops, first-fire, and O_{ctrl}. Error bars are the five-seed sample standard deviation.

RLOR-inspired again sits at one hop ($T_{\mathrm{e2e}} = 1.845\,\mathrm{s}$) because the sink remains in N_{SNR} for live relays; it has no dedicated control plane.

Table IV and Fig. 6 sweep drift magnitude. The hop ratio E-LQTP/LQARP is $1.000\times$ at every v. PDR, delay, and node-exit packet losses match LQARP at every v. Energy per data bit agrees with LQARP through the digits of Table III; across the sweep the largest relative difference is below 3×10^{-4}. O_{ctrl} over the periodic $413.4\,\mathrm{byte/s}$ is $0.05\times$, $0.17\times$, $0.33\times$, $0.52\times$, and $0.62\times$ at 0.5, 1, 2, 3, and $3.5\,\mathrm{m/s}$. First-fire rises with drift, $15\% \to 52\% \to 78\% \to 86\% \to 90\%$. Panel (a) of Fig. 6 is envelope exits, not PDR: the two protocols lose the same relays. Even at $3.5\,\mathrm{m/s}$ E-LQTP stays under the periodic budget.

E. Scenario III: Rankine Eddy

Table V and Fig. 7 repeat the four-protocol comparison under the eddy. E-LQTP again matches LQARP (PDR 0.9989 ± 0.0018, hops $2.050/2.046 = 1.002\times$, delay $4.167\,\mathrm{s}$ versus $4.146\,\mathrm{s}$, energy 1.101×10^{-2} versus $1.099\times10^{-2}\,\mathrm{J/bit}$, node-exit packet losses 0.80 ± 1.30 for both, VOID $= 0$) at $211 \pm 34\,\mathrm{byte/s}$ versus $413.42\,\mathrm{byte/s}$ ($0.51\times$). First-fire is 0.708 ± 0.059; mean exits are 6.40 ± 1.14 of 40, identical across protocols. The leftover 0.2% hop difference is N_{stab} with the trigger on. A separate five-seed run with $\beta_1 = \beta_2 = 1$ together, outside the one-at-a-time sweep of Table VI, produced 3.250 ± 0.141 hops against LQARP's 2.046 ± 0.187 ($1.59\times$). That increase is the scale-mismatched c_{pred} at an $O(1)$ weight. The point is reported only as a diagnostic.

The pair-level trace of Fig. 3 is Scenario III pair $(1, 2)$. The three rising edges at 348, 483, and $512\,\mathrm{s}$ are that pair's HELLOs, not the network-wide count. The network-wide count is O_{ctrl}: the forecast track is allowed to drift, and a HELLO is charged only on a rising edge [15], [16].

F. Reward Ablation

Table VI and Fig. 8 sweep β_1 at $\beta_2 = 0$ and β_2 at $\beta_1 = 0$ on Scenario III. At $\beta_1 = 10^{-6}$ mean hops fall to 2.021 ± 0.158 against LQARP's 2.046 ± 0.187 ($0.988\times$); that dip sits inside LQARP's between-seed band and is not read as a routing gain. Once β_1 leaves the blend-scale range, hops, delay, and energy rise together, reaching $1.14\times$ hops

TABLE III
SCENARIO II AT 2 m/s ($N = 40$, FIVE SEEDS, $\beta_1 = \beta_2 = 0$). ENTRIES WITH $\pm$ ARE THE SAMPLE STANDARD DEVIATION. VOID = 0.

Protocol	PDR	T_{e2e} (s)	E_{bit}	O_{ctrl}	Hops	First-fire
LQARP	0.9992 ± 0.0019	4.438 ± 0.277	$1.156 \pm 0.100 \times 10^{-2}$	0	2.153 ± 0.187	—
E-LQTP	0.9992 ± 0.0019	4.439 ± 0.277	$1.156 \pm 0.100 \times 10^{-2}$	135 ± 23	2.154 ± 0.187	0.782 ± 0.042
RLOR-insp.	1.0000 ± 0	1.845 ± 0.074	5.362×10^{-3}	0	1.000 ± 0.001	—
Periodic	0.9992 ± 0.0019	4.438 ± 0.277	$1.156 \pm 0.100 \times 10^{-2}$	413.42	2.153 ± 0.187	1.000

TABLE IV
SCENARIO II VERSUS DRIFT (LQARP VS. E-LQTP; FIVE SEEDS, $\beta_1 = \beta_2 = 0$). HOP RATIO 1.000× AT EVERY v.

v (m/s)	Protocol	PDR	T_{e2e} (s)	O_{ctrl}	Hops	First-fire
0.5	LQARP	1.000	4.776 ± 0.403	0	2.282 ± 0.186	—
0.5	E-LQTP	1.000	4.776 ± 0.403	20.2 ± 4.2	2.282 ± 0.186	0.146 ± 0.021
1.0	LQARP	0.9995 ± 0.0012	4.752 ± 0.377	0	2.289 ± 0.208	—
1.0	E-LQTP	0.9995 ± 0.0012	4.752 ± 0.377	70.7 ± 6.2	2.289 ± 0.208	0.520 ± 0.031
2.0	LQARP	0.9992 ± 0.0019	4.438 ± 0.277	0	2.153 ± 0.187	—
2.0	E-LQTP	0.9992 ± 0.0019	4.439 ± 0.277	135.1 ± 23.1	2.154 ± 0.187	0.782 ± 0.042
3.0	LQARP	1.000	4.431 ± 0.333	0	2.154 ± 0.203	—
3.0	E-LQTP	1.000	4.433 ± 0.331	213.8 ± 39.1	2.155 ± 0.203	0.857 ± 0.034
3.5	LQARP	1.000	4.200 ± 0.393	0	2.063 ± 0.215	—
3.5	E-LQTP	1.000	4.200 ± 0.393	257.7 ± 46.9	2.063 ± 0.215	0.897 ± 0.028

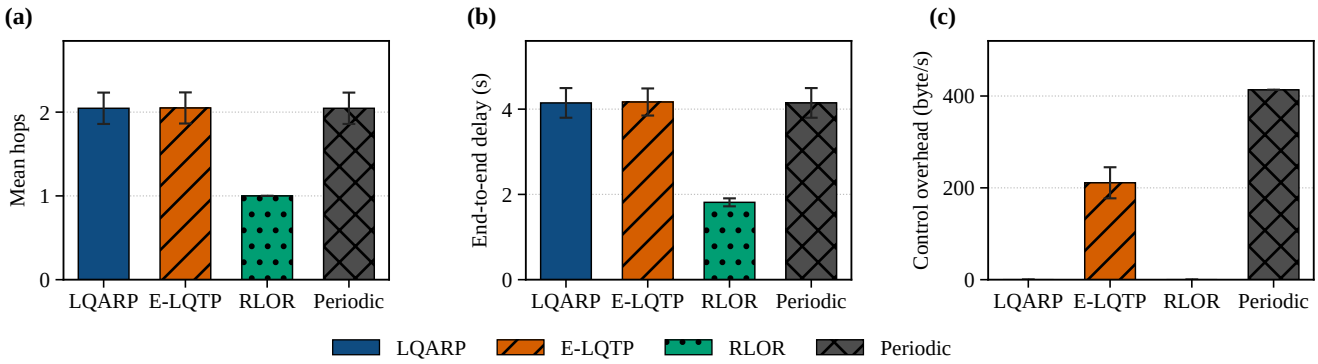


Fig. 7. Scenario III at $V_{\mathrm{eddy}} = 2\,\mathrm{m/s}$: hops, delay, and O_{ctrl}. Error bars are the five-seed sample standard deviation.

TABLE V
SCENARIO III AT $V_{\mathrm{eddy}} = 2\,\mathrm{m/s}$ ($N = 40$, FIVE SEEDS, $\beta_1 = \beta_2 = 0$). ENTRIES WITH $\pm$ ARE THE SAMPLE STANDARD DEVIATION. VOID = 0.

Protocol	PDR	T_{e2e} (s)	E_{bit}	O_{ctrl}	Hops	First-fire
LQARP	0.9989 ± 0.0018	4.146 ± 0.348	$1.099 \pm 0.098 \times 10^{-2}$	0	2.046 ± 0.187	—
E-LQTP	0.9989 ± 0.0018	4.167 ± 0.318	$1.101 \pm 0.097 \times 10^{-2}$	211 ± 34	2.050 ± 0.186	0.708 ± 0.059
RLOR-insp.	1.0000 ± 0	1.814 ± 0.093	5.361×10^{-3}	0	1.000 ± 0	—
Periodic	0.9989 ± 0.0018	4.146 ± 0.348	$1.099 \pm 0.098 \times 10^{-2}$	413.42	2.046 ± 0.187	1.000

and 1.18× delay at $\beta_1 = 10^{-3}$. The β_2 column stays within 0.998×–1.056× LQARP hops with no monotonic trend. No tested weight improves hops, delay, or energy relative to LQARP's between-seed band. A Scenario II-specific β_2 sweep against the sustained AUV–sink range-rate of uniform drift remains future work. The validated operating point is therefore $\beta_1 = \beta_2 = 0$: the control-plane gain is the event trigger, and the reward modifiers are a documented negative result.

G. Discussion Against Recent Journal Protocols

The primary claim is a control-plane claim, not a new data-plane reward. QTAR, SAQR, the hierarchical Q-learning cluster of Yuan *et al.*, ADRP-DQL, CLORP, STGR, and SDN-QLTR all improve forwarding by redesigning the Q-update or the candidate set, and they keep neighbor state current on a timer [10]–[14], [22], [27]. E-LQTP leaves LQARP's reward untouched [9] and charges a HELLO only when the forecast

TABLE VI
SCENARIO III β SWEEPS (FIVE SEEDS). NO TESTED WEIGHT BEATS LQARP'S HOP OR DELAY BAND.

Weight	PDR	T_{e2e} (s)	E_{bit}	Hops
LQARP	0.9989 ± 0.0018	4.146 ± 0.348	$1.099 \pm 0.098 \times 10^{-2}$	2.046 ± 0.187
β_1 sweep ($\beta_2 = 0$)				
0	0.9989 ± 0.0018	4.167 ± 0.318	$1.101 \pm 0.097 \times 10^{-2}$	2.050 ± 0.186
10^{-7}	0.9986 ± 0.0017	4.159 ± 0.324	$1.098 \pm 0.092 \times 10^{-2}$	2.043 ± 0.175
10^{-6}	1.0000 ± 0	4.090 ± 0.279	$1.083 \pm 0.085 \times 10^{-2}$	2.021 ± 0.158
10^{-5}	1.0000 ± 0	4.301 ± 0.399	$1.132 \pm 0.103 \times 10^{-2}$	2.111 ± 0.192
10^{-4}	0.9976 ± 0.0040	4.523 ± 0.082	$1.185 \pm 0.056 \times 10^{-2}$	2.194 ± 0.097
10^{-3}	0.9980 ± 0.0027	4.897 ± 0.351	$1.253 \pm 0.099 \times 10^{-2}$	2.330 ± 0.188
β_2 sweep ($\beta_1 = 0$)				
0	0.9989 ± 0.0018	4.167 ± 0.318	$1.101 \pm 0.097 \times 10^{-2}$	2.050 ± 0.186
0.25	0.9989 ± 0.0018	4.140 ± 0.304	$1.097 \pm 0.103 \times 10^{-2}$	2.042 ± 0.197
0.50	0.9989 ± 0.0018	4.214 ± 0.349	$1.115 \pm 0.106 \times 10^{-2}$	2.076 ± 0.204
1.00	0.9989 ± 0.0018	4.405 ± 0.415	$1.160 \pm 0.126 \times 10^{-2}$	2.161 ± 0.240
2.00	0.9989 ± 0.0018	4.221 ± 0.293	$1.117 \pm 0.090 \times 10^{-2}$	2.080 ± 0.171

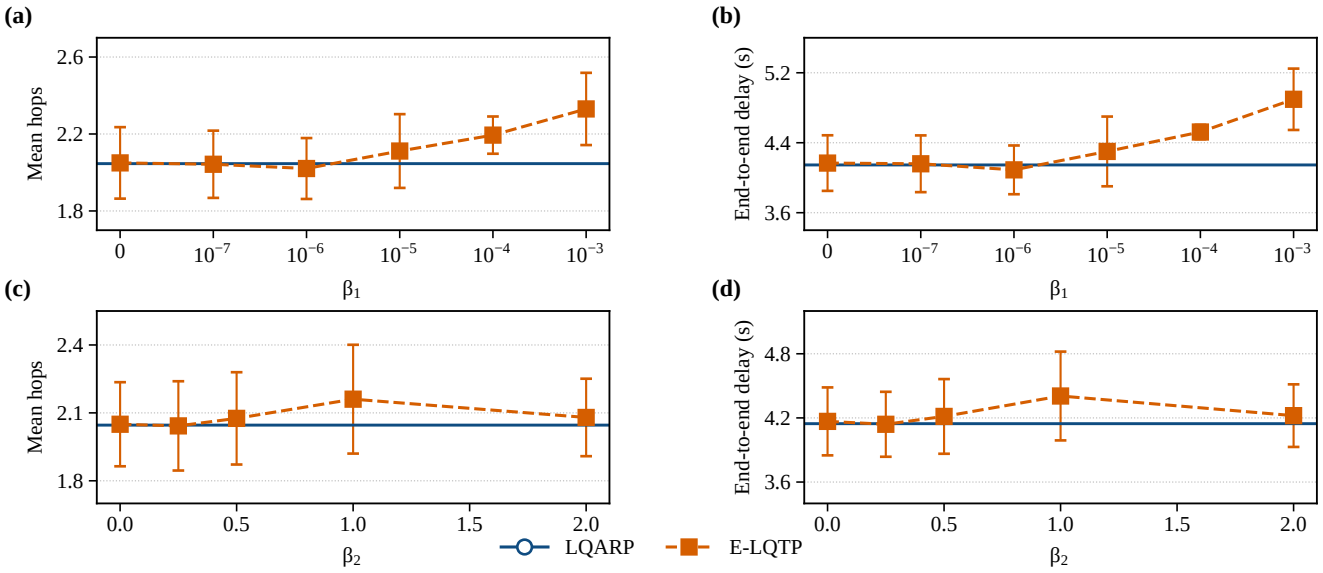


Fig. 8. Scenario III reward sweep. Solid line: LQARP. Error bars are the five-seed sample standard deviation. No tested β improves hops or delay.

track diverges from the per-tick channel query. Scenario II and Scenario III match LQARP's hops and delay to three figures because both routers read the same live pose and the validated weights are $\beta_1 = \beta_2 = 0$, while O_{ctrl} falls to $0.33\times$ and $0.51\times$ the $5\,\mathrm{s}$ grid used as the periodic baseline in this campaign. Those factors are the timing ablation under the shared pose of Section III-E. QTAR, SAQR, and STGR are cited as members of that timer class [10]–[12]. They were not reimplemented on this site, so the factors above are not a claim that their published overhead equals $413.42\,\mathrm{byte/s}$. QLAR is the design that already adapts a HELLO period, and it does so from predicted link lifetime rather than from (13) [28].

Trust-aware and clustered opportunistic designs (He *et al.*, RCRP, FHOEEC, LSLPR) and the information-centric push/pull family (push-hybrid UNDN, RECAR, Wang–Lu–Gao) likewise treat the beacon as a security, void, or naming event [20], [21], [26], [33], [35]–[38]. None of those events is the forecast-versus-truth SNR residual of Section III, so their control traffic does not automatically collapse when relative geometry is temporarily stable. AUV collection and path-planning DQNs [6]–[8], [29] move the vehicle rather than the HELLO timer; they are complementary to E-LQTP, not substitutes. The same holds for acoustic–radio hybrid hops [30], multi-objective and cross-layer schedules [31], [32], MARV's value-of-information split [25], GMM–HMM–LSTM adaptation and trust guidelines [5], [34], and the Zhoushan multipath sea trial [39]: each changes the forwarding objective or the physical path, not the HELLO cadence.

The event-triggered Kalman literature [15]–[18], [40], [41] already showed that a predicted information pair can replace a periodic sample. E-LQTP reuses that idea for a routing HELLO on a per-tick Bellhop query. The $2\,\mathrm{dB}$ cap is this paper's choice, placed below the LOON21 Kalman mean absolute errors 4.81, 3.47, and $3.24\,\mathrm{dB}$ [19]. The β ablation then shows that folding the same innovation into c_{pred} and c_{drift} does not buy a data-plane gain on this grid—a scoped negative result relative to the reward-heavy Q-learning family above.

V. Conclusion

THIS article addressed the control-plane cost of keeping link-quality state accurate in a mobile underwater mesh. E-LQTP adds a dual-track extended Kalman filter to LQARP so that a HELLO fires only when a forecast that receives no measurement update diverges from the per-tick channel query. With the trigger on and $\beta_1 = \beta_2 = 0$, E-LQTP matches LQARP's packet delivery ratio, hop count, end-to-end delay, and energy per data bit, while cutting control traffic to $0.33\times$ a $5\,\mathrm{s}$ periodic grid under $2\,\mathrm{m/s}$ drift and to $0.51\times$ under the Rankine eddy. The campaign counts those control bytes in O_{ctrl}. Reception is the SNR gate of Section III-F, and HELLO energy is left out of residual energy, so the reported saving is the byte count on the shared live pose. The additive reward terms do not improve hops or delay; that negative result is reported rather than tuned away, isolating the gain to the event-triggered mechanism itself.

A relative-form c_{pred} and a Scenario II-only β_2 sweep against sustained AUV–sink range-rate remain future work, together with a neighbor table that ages between velocity copies and a non-ground-truth channel-estimation pipeline for the staleness test.

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REFERENCES

[1] S. Gupta and N. P. Singh, "Underwater wireless sensor networks: a review of routing protocols, taxonomy, and future directions," *The Journal of Supercomputing*, vol. 80, no. 4, pp. 5163–5196, 2024.

[2] K. K. Gola and S. Arya, "Underwater acoustic sensor networks: Taxonomy on applications, architectures, localization methods, deployment techniques, routing techniques, and threats: A systematic review," *Concurrency and Computation: Practice and Experience*, vol. 35, no. 23, p. e7815, 2023.

[3] K. Saleem, L. Wang, and S. Bharany, "Survey of AI-driven routing protocols in underwater acoustic networks for enhanced communication efficiency," *Ocean Engineering*, vol. 314, p. 119606, Dec. 2024.

[4] J. Jiang, G. Han, and C. Lin, "A survey on opportunistic routing protocols in the Internet of Underwater Things," *Computer Networks*, vol. 225, p. 109658, Apr. 2023.

[5] R. Zhu, A. Boukerche, L. Long, and Q. Yang, "Design guidelines on trust management for underwater wireless sensor networks," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 4, pp. 2547–2576, 2024.

[6] J. Wang, S. Liu, W. Shi, G. Han, and S. Yan, "A multi-AUV collaborative ocean data collection method based on LG-DQN and data value," *IEEE Internet of Things Journal*, vol. 11, no. 5, pp. 9086–9106, Mar. 2024.

[7] Z. Liu, J. Hou, D. Ning, C. Zhou, G. Liang, and F. Zhang, "Improving deep Q network based on marketing psychology for AUV path planning in unknown marine environments," *IEEE Internet of Things Journal*, vol. 12, no. 5, pp. 5476–5487, Mar. 2025.

[8] Y. Lu and Y. Zuo, "Multiobjective optimization of path planning and communication capacity based on DQN with weighted prioritized experience replay," *IEEE Internet of Things Journal*, vol. 12, no. 24, pp. 53 262–53 273, Dec. 2025.

[9] X. Liu, F. Zhou, X. Ji, X. Qu, and G. Qiao, "Link quality-aware routing protocol for underwater wireless sensor networks: A reinforcement learning approach," *IEEE Internet of Things Journal*, vol. 13, no. 17, pp. 39 561–39 576, Sep. 2026.

[10] C. Xu, S. Song, X. Wu, G. Han, M. Pan, G. Xu, and J.-H. Cui, "A high reliable routing protocol based on spatial-temporal graph model for multiple unmanned underwater vehicles network," *IEEE Transactions on Mobile Computing*, vol. 24, no. 5, pp. 4434–4450, May 2025.

[11] C. S. Nandyala, H.-W. Kim, and H.-S. Cho, "QTAR: A Q-learning-based topology-aware routing protocol for underwater wireless sensor networks," *Computer Networks*, vol. 222, p. 109562, Feb. 2023.

[12] S. Zhang, H. Chen, and L. Xie, "Service-aware Q-learning-based routing protocol in the underwater acoustic sensor network," *Computer Networks*, vol. 257, p. 110986, Feb. 2025.

[13] S. Liu, J. Wang, W. Shi, G. Han, S. Yan, and J. Li, "CLORP: Cross-layer opportunistic routing protocol for underwater sensor networks based on multiagent reinforcement learning," *IEEE Sensors Journal*, vol. 24, no. 10, pp. 17 243–17 258, May 2024.

[14] A. S. Mohanraju and A. Lokam, "ADRP-DQL: An adaptive distributed routing protocol for underwater acoustic sensor networks using deep Q-learning," *Ad Hoc Networks*, vol. 167, p. 103692, Feb. 2025.

[15] G. Li, C. Yang, G. Gao, X. Zhang, E. Wang, W. Wang, and D. Niyato, "Distributed state estimation with two event-triggered communication strategies via Internet of Underwater Things," *IEEE Internet of Things Journal*, vol. 13, no. 2, pp. 2345–2359, Jan. 2026.

[16] A. Priel and D. Zelazo, "Event-triggered consensus Kalman filtering for time-varying networks and intermittent observations," *International Journal of Robust and Nonlinear Control*, vol. 33, no. 13, pp. 7430–7451, 2023.

[17] E. Sebastián, E. Montijano, and C. Sagués, "ECO-DKF: Event-triggered and certifiable optimal distributed Kalman filter under unknown correlations," *IEEE Transactions on Automatic Control*, vol. 69, no. 4, pp. 2613–2620, Apr. 2024.

[18] D. Deng and J. Xiong, "Stochastic event-triggered remote state estimation over Gaussian channels without knowing triggering decisions: A Bayesian inference approach," *Automatica*, vol. 152, p. 110951, Jun. 2023.

[19] F. Busacca, L. Galluccio, S. Palazzo, and A. Panebianco, "A comparative analysis of predictive channel models for real shallow water environments," *Computer Networks*, vol. 250, p. 110557, Aug. 2024.

[20] R. Wang, Y. Liu, J. Liu, H. Yin, and Z. Guo, "Push-hybrid data forwarding in underwater named data networking," *IEEE Internet of Things Journal*, vol. 11, no. 24, pp. 40 319–40 330, Dec. 2024.

[21] Y. Li, Y. Liu, H. Yin, Z. Guo, and Y. Wang, "RECAR: Robust and efficient collision-avoiding routing for 3D underwater named data networking," *Journal of Network and Computer Applications*, vol. 230, p. 103940, Oct. 2024.

[22] Y. Yuan, M. Liu, X. Zhuo, Y. Wei, X. Tu, and F. Qu, "A Q-learning-based hierarchical routing protocol with unequal clustering for underwater acoustic sensor networks," *IEEE Sensors Journal*, vol. 23, no. 6, pp. 6312–6325, Mar. 2023.

[23] C. Wang, X. Shen, H. Wang, H. Zhang, and H. Mei, "Reinforcement learning-based opportunistic routing protocol using depth information for energy-efficient underwater wireless sensor networks," *IEEE Sensors Journal*, vol. 23, no. 15, pp. 17 771–17 783, Aug. 2023.

[24] C. Wang, X. Shen, H. Wang, W. Xie, H. Mei, and H. Zhang, "Q-learning-based routing protocol with accelerating convergence for underwater wireless sensor networks," *IEEE Sensors Journal*, vol. 24, no. 7, pp. 11 562–11 573, Apr. 2024.

[25] C. Wang, X. Shen, H. Wang, W. Xie, H. Zhang, and H. Mei, "Multi-agent reinforcement learning-based routing protocol for underwater wireless sensor networks with value of information," *IEEE Sensors Journal*, vol. 24, no. 5, pp. 7042–7054, Mar. 2024.

[26] Y. He, G. Han, Y. Hou, and C. Lin, "Environment-tolerant trust opportunity routing based on reinforcement learning for Internet of Underwater Things," *IEEE Transactions on Mobile Computing*, vol. 24, no. 7, pp. 6348–6360, Jul. 2025.

[27] F. Li, G. Han, C. Lin, F. Zhang, and C. Sun, "SDN-QLTR: Q-learning-assisted trust routing scheme for SDN-based underwater acoustic sensor networks," *IEEE Internet of Things Journal*, vol. 11, no. 6, pp. 10 682–10 694, Mar. 2024.

[28] X. Li, Y. Wu, M. Zhu, and J. Ren, "A Q-learning-based link-aware routing protocol for underwater wireless sensor networks," *Journal of Marine Science and Engineering*, vol. 13, no. 12, p. 2374, 2025.

[29] C. S. Nandyala and H.-S. Cho, "AUV-aided isolated sub-network prevention for reliable data collection by underwater wireless sensor networks," *Computer Networks*, vol. 262, p. 111154, May 2025.

[30] Z. Jiang, Q. Guan, F. Chen, N. Wei, F. Ji, and H. Yu, "Opportunistic hybrid routing protocol for acoustic-radio cooperative networks," *IEEE Internet of Things Journal*, vol. 10, no. 21, pp. 19 014–19 026, Nov. 2023.

[31] Y. Yuan, X. Zhuo, Y. Wei, and F. Qu, "Multiobjective routing optimization to support IoUT applications in UWSNs," *IEEE Internet of Things Journal*, vol. 11, no. 16, pp. 27 664–27 675, Aug. 2024.

[32] Y. Yuan, X. Zhuo, M. Liu, Y. Wei, and F. Qu, "A centralized cross-layer protocol for joint power control, link scheduling, and routing in UWSNs," *IEEE Internet of Things Journal*, vol. 11, no. 7, pp. 12 823–12 833, Apr. 2024.

[33] R. Zhu, A. Boukerche, Y. Chen, and Q. Yang, "A reliable cluster-based opportunistic routing protocol for underwater wireless sensor networks," *Computer Networks*, vol. 251, p. 110622, Sep. 2024.

[34] R. Zhu, A. Boukerche, and Q. Yang, "An efficient secure and adaptive routing protocol based on GMM-HMM-LSTM for Internet of Underwater Things," *IEEE Internet of Things Journal*, vol. 11, no. 9, pp. 16 491–16 504, May 2024.

[35] S.-W. Lee, M. Alhussein, K. Aurangzeb, M. S. Yousefpoor, E. Yousefpoor, and M. Hosseinzadeh, "A fire hawk optimizer-based energy-efficient clustering scheme in underwater acoustic sensor networks (UASNs)," *Ad Hoc Networks*, vol. 176, p. 103889, Sep. 2025.

[36] X. Wang, Y. Lu, and D. Gao, "Reliable data forwarding for information-centric underwater IoT," *IEEE Transactions on Reliability*, vol. 74, no. 3, pp. 4222–4229, Sep. 2025.

[37] R. Zhao, W. Xie, X. Shen, C. Wang, and H. Wang, "Time-reversal cross-layer opportunistic routing protocol for reliable Internet of Underwater Things," *IEEE Internet of Things Journal*, vol. 13, no. 6, pp. 11 330–11 344, Mar. 2026.

[38] X. Tian, X. Du, L. Wang, L. Zhao, and D. Han, "LSLPR: A layering and source-location-privacy-based routing protocol for underwater acoustic sensor networks," *IEEE Sensors Journal*, vol. 23, no. 19, pp. 23 676–23 691, Oct. 2023.

[39] Y. Yuan, T. Lou, S. Yang, Y. Yuan, X. Zhuo, Y. Wei, and F. Qu, "Source-based disjoint multipath routing protocol for underwater acoustic networks: Protocol design and sea trials," *IEEE Sensors Journal*, vol. 24, no. 22, pp. 38 061–38 070, Nov. 2024.

[40] D. Deng and J. Xiong, "Kalman-like filter for event-triggered remote state estimation over an additive noise channel," *IEEE Control Systems Letters*, vol. 8, pp. 205–210, 2024.

[41] Y. Zhong, L. Huang, Y. Mo, D. Shi, and L. Shi, "Event-triggered multi-sensor scheduling for remote state estimation over packet-dropping networks," *IEEE Transactions on Signal Processing*, vol. 72, pp. 5036–5047, 2024.